Spring 2017 **PHY102: Introduction to Physics-II** **26/04/17**

Tutorial – 15

Department of Physics **Shiv Nadar University**

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1. Show that the standing wave $f(z,t) = A\sin(kz)\cos(kvt)$ satisfies the wave equation, and express it as the sum of a wave traveling to the left and a wave traveling to the right.

2. Write down the real electric and magnetic fields for a monochromatic wave of amplitude E_0, frequency ω, and phase angle zero that is (a) traveling in the negative x direction and polarized in the z direction; (b) traveling in the direction form the origin to the point (1, 1, 1), with polarization parallel to the x-z plane.

3. The intensity of sunlight hitting the earth is about 1300 W/m^2. If sunlight strikes a perfect absorber, what pressure does it exert? How about a perfect reflector? What fraction of atmospheric pressure does this amount to?

4. Consider the case of a 2-slit diffraction where $b = 8.8 \times 10^{-3}$ cm (width of each slit), $d = 7.0 \times 10^{-2}$ cm (separation between slits). Plane waves of wavelength $\lambda = 6.328 \times 10^{-5}$cm are incident on both the slit normally. How many interference minima will occur between the two diffraction minima on either side of the central maximum? Calculate the fringe width of the interference pattern if the screen is placed at a distance of 15 ft from the lens.

5. A parallel beam of light is incident normally on a narrow slit of width 0.2 mm. The Fraunhofer diffraction pattern is observed on a screen which is placed at the focal plane of a convex lens whose focal length is 20 cm. Calculate the distance between the first two minima and the first two maxima on the screen. Assume that $\lambda = 5 \times 10^{-5}$cm and that the lens is placed very close to the slit.